

The full-merge lead is a connected-graph cumulant (Theorems G and F)

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WIP commit: d32be4d of `grandpa-rick/work-in-progress`. This is the latest commit touching the source file `proofs/2026-10-05-day221-lead-cumulant.md`, and this PDF is a faithful transcription of that file.

Registry grades: nodes `conjG` and `conjF` are *proved*. No one else has read them yet, and **novelty has not been searched** (see §8).

Provenance note. The proof file committed in WIP 5b73f01 was accidentally truncated to about 1.2 KB: the header, a list of scripts and §8 survived, and §§1–7 were lost. This PDF is the restored version, which I re-derived from scratch in commit d32be4d from the session outline and the Day 220 proofs of Theorems B and W. That makes it a second derivation, not the original text. The text of the Day 220 Theorem W cold re-check (old §6) was also lost and has *not* been restored. Theorem W is graded proved on its own Day 220 proof (`proofs/2026-10-03-day220-s1-carre-du-champ.md` §5b), but its independent re-read is still owed.

0. Setup (from Day 220 §4)

Let $\lambda = (\lambda_1, \dots, \lambda_\ell) \vdash n$, with parts indexed by positions $[\ell]$. Let μ be a coarsening of λ , and put $m = \ell(\lambda) - \ell(\mu)$. Define $\text{Lead}_{\lambda\mu}(t) := [(s-1)^m] c_{\lambda\mu}$. By Theorem B (Day 220; proved, and it does not use (N)),

$$\text{Lead}_{\lambda\mu} = \sum_{\text{tight histories } H \text{ ending at } \mu} \prod_{\text{merge steps}} W_k(J).$$

A history processes positions $\ell, \ell-1, \dots, 1$ in turn. At position i , where $k = \lambda_i$, it either opens a new block or merges with $p \geq 1$ existing blocks of sizes $J = (j_1, \dots, j_p)$, producing one block of size $k + |J|$. By Theorem W (Day 220 §5b),

$$W_k(J) = (-1)^p [k + |J|]_t \frac{\prod_{j \in J} [k]_{t^j}}{[k]_t}.$$

Opening a block has weight 1, which is also $W_k(\emptyset)$.

1. Lemma 1

Lemma 1 (histories are increasing forests). *Histories correspond bijectively to increasing forests on $[\ell]$, and the final blocks are the trees.*

Proof. When position i merges with existing blocks, make i the parent of each of those blocks' current roots. Every existing block was built only from positions $> i$, so every edge goes from a smaller label to a larger one. A history therefore gives an increasing forest on $[\ell]$, and its trees are the final blocks. Conversely, an increasing forest determines the history: at step i , merge exactly the subtrees rooted at the children of i . Those subtrees are complete by then, because all their labels are $> i$. So the correspondence is a bijection. Histories ending in a single block are increasing trees, and there are $(\ell-1)!$ of them. A tight history needs $\sum p = m$, which is automatic: each edge is one unit of p , and a forest with $\ell(\mu)$ trees has $\ell - \ell(\mu)$ edges. $\square$

2. Lemma 2

Lemma 2 (Theorem W telescopes). *For a vertex v , let N_v be the λ -mass of its subtree. For an increasing tree with root mass n ,*

$$\prod_v W = \frac{1 - t^n}{\prod_i (1 - t^{\lambda_i})} \prod_{\text{edges } v \rightarrow c} (t^{\lambda_v N_c} - 1).$$

Proof. The merge at v has $k = \lambda_v$ and $J = (N_c)_c$ child of v , so $k + |J| = N_v$. Then

$$\prod_v W = (-1)^{\#\text{edges}} \prod_v \frac{[N_v]_t}{[\lambda_v]_t} \cdot \prod_{\text{edges } v \rightarrow c} [\lambda_v]_{t^{N_c}}.$$

Here $[N_v]_t / [\lambda_v]_t = (1 - t^{N_v}) / (1 - t^{\lambda_v})$ and $[\lambda_v]_{t^{N_c}} = (1 - t^{\lambda_v N_c}) / (1 - t^{N_c})$. Every non-root c appears exactly once as a child, so its factor $(1 - t^{N_c})$ cancels, leaving only the root's $(1 - t^n)$. The claim follows from $(-1)^{\#\text{edges}} \prod (1 - x) = \prod (x - 1)$. Leaves have $N_v = \lambda_v$, which gives the factor 1, consistent with opening a block. $\square$

3. Lemma 3

Lemma 3 (tree-graph identity, generic weights). *For indeterminates $x_{ij} = x_{ji}$ ($i \neq j$ in $[\ell]$),*

$$\sum_{\text{increasing trees } T \text{ on } [\ell]} \prod_{\text{edges } v \rightarrow c} \left(\prod_{j \in T_c} x_{vj} - 1 \right) = \sum_{\text{connected graphs } G \text{ on } [\ell]} \prod_{e \in G} (x_e - 1),$$

where T_c is the vertex set of the subtree at c .

Proof. Map a connected graph G to a tree recursively. The root is $\min V = 1$. The components $C_1, \dots, C_r$ of $G - 1$ become the child subtrees, with their minima as children, and we recurse on each $G[C_a]$. Each C_a is connected, so the recursion is defined. Because every child is the minimum of its subtree, the result is increasing. Every increasing tree arises this way, since its child subtrees are nested set partitions with minimum roots.

Fix T and sum over the graphs G that map to it. At each vertex v , the edges from v into the child set T_c form an arbitrary *nonempty* subset $S \subseteq T_c$. It must be nonempty because T_c has to be attached to v . It is otherwise free, because the edges from v to T_c do not change the component structure inside T_c . There are no edges between different child sets, since they are different components of $G[T_v] - v$. The edges inside T_c are accounted for by the recursion. Hence the fibre sum factors, and for each edge $v \rightarrow c$

$$\sum_{\emptyset \neq S \subseteq T_c} \prod_{j \in S} (x_{vj} - 1) = \prod_{j \in T_c} x_{vj} - 1. \quad \square$$

Specialization. Setting $x_{ij} = t^{\lambda_i \lambda_j}$ gives $\prod_{j \in T_c} x_{vj} = t^{\lambda_v N_c}$.

4. Theorem G

Theorem G. *For $\lambda \vdash n$ with $\ell \geq 2$,*

$$\text{Lead}_{\lambda, (n)}(t) = \frac{1 - t^n}{\prod_i (1 - t^{\lambda_i})} K_\lambda(t), \quad K_\lambda := \sum_{G \text{ connected on } [\ell]} \prod_{ij \in G} (t^{\lambda_i \lambda_j} - 1).$$

Proof. Apply Lemma 1 restricted to trees, then Lemma 2, then Lemma 3 with the specialization. $\square$

By the exponential formula, K_λ is the connected part (joint cumulant) of $t^{e_2(\lambda)} = \prod_{i < j} (1 + (t^{\lambda_i \lambda_j} - 1))$, i.e. the logarithm of a group-like element in the graph Hopf algebra.

Corollaries.

- $\lambda = 1^n$: the prefactor is $[n]_t / (1 - t)^{n-1}$, and $K = \sum_{\text{conn}} (t - 1)^{|E|} = (t - 1)^{n-1} I_n(t)$, where I_n is the Mallows–Riordan tree inversion enumerator. So $\text{Lead} = (-1)^{n-1} [n]_t I_n(t)$.
- $t = 0$: every edge factor is -1 , so $\sum_{\text{conn}} (-1)^{|E|} = (-1)^{\ell-1} (\ell - 1)!$. Alternatively, count the trees directly.
- $t \rightarrow 1$: with $\varepsilon = 1 - t$, the tree form gives $\text{Lead} \rightarrow (-1)^{\ell-1} \frac{n}{\prod \lambda_i} \sum_T \prod_{\text{edges } v \rightarrow c} \lambda_v N_c$. The claimed value $(-1)^{\ell-1} n^{\ell-1}$ is equivalent to $\sum_T \prod \lambda_v N_c = n^{\ell-2} \prod \lambda_i$, a weighted Cayley identity. It checks at $\ell = 2$ (giving $\lambda_1 \lambda_2$). The general case is a claim from the PROVE session that is *not re-derived here*; please check it before citing.
- Sign: for $0 < t < 1$ the prefactor is positive and each of the $\ell - 1$ edge factors is negative. For $t > 1$ the prefactor has sign $(-1)^{\ell-1}$ and the edge factors are positive. Either way $\text{sgn Lead} = (-1)^{\ell-1}$ for all $t > 0$, $t \neq 1$.

5. Theorem F

Theorem F. For μ a coarsening of λ ,

$$\text{Lead}_{\lambda\mu} = \sum_{\substack{\text{set partitions } \pi \text{ of } [\ell] \\ \text{with sorted block sums}=\mu}} \prod_{B \in \pi} \text{Lead}_{\lambda_B, (|\lambda_B|)},$$

where a singleton block contributes 1. There is no interleaving factor.

Proof. By Lemma 1, increasing forests correspond to a set partition together with an increasing tree on each block. The weight is a product over the trees, and by Lemma 2 each tree’s weight depends only on its own parts. Sum. $\square$

6. Theorem W sober re-check — lost

The original §6 was a line-by-line re-derivation of Day 220 §5b: signs, top-symbol count, Macdonald III.7 Ex. 2 at $n = 2$, III.2 Ex. 1 multiplicities, and Cauchy. It was lost in the truncation. The session log records “no gap”, but the text no longer exists, so the re-check should be treated as unrecorded.

7. Computation (scripts survive)

- `scripts/day221/tree_identity.py` $\rightarrow$ `tree_identity.log`: Lemma 3 with random rational generic x_{ij} , 15/15 ($\ell = 2..6$). The history sum (Theorem W weights) equals the Theorem G closed form in exact symbolic t (SymPy), **37/37 for all $\lambda \vdash n \leq 7$ with $\ell \geq 2$.**
- `scripts/day221/tree_identity_n8.py` $\rightarrow$ `tree_identity_n8.log`: history = telescoped tree form (Lemma 2) = Theorem G, in exact rationals at $t \in \{3/5, 7/3, -2, 2\}$, for all $\lambda \vdash n \leq 8$ with $2 \leq \ell \leq 6$: **220/220.**
- Earlier wake data: G 58/58 ($n \leq 8$); F 123/123 ($n \leq 7$); subset-engine agreement 32/32 and 132/132.

8. Gaps

- G and F have no gaps beyond their premises: Theorem B (proved, does not use (N)), Theorem W (proved; the re-check text was lost, see §6) and (KF) (proved, Day 216b).
- The $t = 1$ corollary value is not re-derived here.
- Novelty has not been searched. Lemma 3 looks classical (a tree-graph / Penrose-type decomposition; Gessel–Wang / Mallows–Riordan at uniform weights), so the new content would be identifying the leading coefficient of $\star$ with it. Search targets: “connected graphs” with weight $t^{ab} - 1$ and the prefactor $(1 - t^n) / \prod (1 - t^{\lambda_i})$; coloured inversion enumerators; q -Cayley.